

PHYS 661 Fall 2024

Problem set 3, due October 3 2024. Each problem is 10 pts unless otherwise stated. Submit your homework either (1) via Brightspace or by (2) emailing it directly to the TA at gbowling@purdue.edu.

1 Variational principle (25 pts)

In this problem you will learn the variational principle (see Sakurai Sec. 5.4 for more details) and apply it to the case of a double-well potential. The basic idea is that a guess $|\psi\rangle$ for the ground state of a Hamiltonian H satisfies $\langle\psi|H|\psi\rangle \geq E_0$, where E_0 is the exact ground state energy and $|\psi\rangle$ is assumed to be normalized. We can try to minimize $\langle\psi|H|\psi\rangle$ (and improve our bound on E_0) by taking derivative w.r.t. a parameter β in the guess state: $\partial_\beta \langle\psi(\beta)|H|\psi(\beta)\rangle = 0$.

1. (10 pts) Consider a 1D Harmonic oscillator Hamiltonian $H = \frac{p_x^2}{2m} + \frac{1}{2}m\omega_0^2 x^2$. Find the energy $E(\beta) = \langle\psi|H|\psi\rangle$ and its minimum value $E(\beta_*)$ for the following variational wave functions. Interpret your results.

(a) $\psi_a(x, \beta) = \left(\frac{2\beta}{\pi}\right)^{1/4} \exp(-\beta x^2)$

(b) $\psi_b(x, \beta) = \begin{cases} \left(\frac{15}{16}\right)^{1/2} \beta^{1/4} (1 - \beta x^2) & |x| < \beta^{-1/2} \\ 0 & |x| > \beta^{-1/2} \end{cases}$

2. (15 pts) Let us consider the double well potential with Hamiltonian $H = \frac{p_x^2}{2m} + \frac{1}{2}m\frac{1}{4}\omega_0^2 a^2 \left(\frac{x^2}{a^2} - 1\right)^2$.

- (a) (5 pts) Using $\psi_1 = \left(\frac{2\beta}{\pi}\right)^{1/4} \exp(-\beta(x-a)^2)$, find $E_1(\beta) = \langle\psi_1|H|\psi_1\rangle$. Minimizing $E_1(\beta)$ is in general cumbersome, but show that for large a , the minimum is given by $\beta_* = \frac{m\omega_0}{2\hbar}$. Find $E_1(\beta_*)$ and comment on your result.

- (b) (10 pts) Use now instead $\psi_2 = A[\exp(-\beta(x-a)^2) + \exp(-\beta(x+a)^2)]$. Find the normalization constant A . Evaluate $E_2(\beta) = \langle\psi_2|H|\psi_2\rangle$. Minimizing E_2 in this case is hopeless to do analytically. However, by using the above value $\beta_* = \frac{m\omega_0}{2\hbar}$, find the difference $E_2(\beta_*) - E_1(\beta_*)$ in the limit of large a and comment on your result.

2 WKB approximation

Consider a 1D potential $V(x) = \begin{cases} \infty & x < 0 \\ \frac{1}{2}m\omega_0^2 x^2 & x > 0 \end{cases}$, i.e., a harmonic trap interrupted by a hard wall. By using the WKB approximation, find the bound state energies.

3 Wave function normalization in perturbation theory

In the lectures we chose the perturbed state $|n\rangle$ (with energy E_n) to be such that $\langle n^{(0)}|n\rangle = 1$. Let us define a *normalized* perturbed state $|n\rangle_N = Z_n^{1/2}|n\rangle$ (see lecture notes Sec. 2.1.1) which satisfies ${}_N\langle n|n\rangle_N = 1$. Show that

$$Z_n = \frac{\partial E_n}{\partial E_n^{(0)}}, \quad (1)$$

where $E_n^{(0)}$ is the energy of the unperturbed state $|n^{(0)}\rangle$.